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| **Fiscal Year:** | FY 2005 | **Task Last Updated:** | FY 03/31/2006 |
| **PI Name:** | Turner, Ronald E Ph.D. | | |
| **Project Title:** | Solar Particle Event Observations with MARIE, and Correlations with Earth and Near-Earth Instruments | | |
| **Division Name:** | Human Research | | |
| **Program/Discipline:** | HUMAN RESEARCH | | |
| **Program/Discipline--Element/Subdiscipline:** | HUMAN RESEARCH--Radiation health | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | None | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | None | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
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| **PI Organization Type:** | INDUSTRY | **Phone:** | 703-416-3264 |
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| **City:** | Arlington | **State:** | VA |
| **Zip Code:** | 22206 | **Congressional District:** | 4 |
| **Comments:** | | | |
| **Project Type:** | Ground | **Solicitation / Funding Source:** | NOT AVAILABLE |
| **Start Date:** | 04/22/2004 | **End Date:** | 07/31/2006 |
| **No. of Post Docs:** | | **No. of PhD Degrees:** | |
| **No. of PhD Candidates:** | | **No. of Master' Degrees:** | |
| **No. of Master's Candidates:** | | **No. of Bachelor's Degrees:** | |
| **No. of Bachelor's Candidates:** | | **Monitoring Center:** | NASA JSC |
| **Contact Monitor:** | | **Contact Phone:** | |
| **Contact Email:** | | | |
| **Flight Program:** | | | |
| **Flight Assignment:** | NOTE: received NCE to 7/31/2006 per J. Dardano/JSC (5/06) | | |
| **Key Personnel Changes/Previous PI:** | | | |
| **COI Name (Institution):** | | | |
| **Grant/Contract No.:** | NAG9-1427 | | |
| **Performance Goal No.:** | | | |
| **Performance Goal Text:** | | | |
| **Task Description:** | The Odyssey mission presents an exciting opportunity for advances in radiation risk management of a future human mission to Mars. A major objective of this mission is to characterize the Martian radiation environment to support future human missions to Mars. In addition, the MARIE instrument, designed to measure the energetic particle flux at Mars, can be used to provide multipoint observations of solar particle events (SPEs) in the critical region of the heliosphere needed to reduce the in-flight radiation risk to a future Mars-bound crew.<br>MARIE collected data in Mars orbit from March 2002 to the end of October 2003. While efforts continue to revive the instrument, the data collected to date have already been very valuable and offer insight into the heliospheric evolution and structure of solar particle events. This work will examine MARIE data collected during SPEs, extract critical information including timing and magnitude of event onset, peak, possible shock-enhanced peaks, and subsequent decay. | | |

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| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | |
| **Task Progress:** | The primary goal of this study was to identify and catalog the time of SPE onset, the evolution of the SPE, the identification of shock-enhanced peaks, and the duration of the event. This information has been correlated with similar data taken near Earth during the event by the NOAA GOES spacecraft. It has also been correlated with published observations of the solar source event, generally as observed by instruments on SOHO, but also as observed by the NOAA X-ray Imager. A catalogue and visual summary of SPEs observed by MARIE and GOES was produced. A notional model of the time evolution of an SPE was created and adapted to be consistent with the data. During the second year, a generalized model of an SPE was developed. While preliminary work showed that such a model produces a representation of an SPE that is consistent with observations, more work is needed to fit the model more rigorously with the data. A significant challenge is to identify and characterize the source solar activity, particularly since in many cases multiple solar events occur in quick succession, each contributing to some degree to the elevated flux levels.<br>The collected observations have been shared with the solar physics and life science community through publications and participation at American Geophysical Union Conferences, the annual NSF-funded Solar and Heliospheric and Interplanetary Environments (SHINE) workshops, meetings of the Mars Exploration Analysis Group (MEPAG) and at appropriate Life Sciences conferences, such as the annual Space Radiation Health Investigators workshops.<br><br>The primary benefit of this work has been to ensure that the valuable data collected by MARIE is disseminated in useful form to the scientists in other disciplines that may otherwise not have access to the data or even be aware of its existence. The exchange of information with the space physics, space flight, and life science communities will enhance the continued development of SPE models and forecast tools. This, in turn, will lead to a better understanding of the three-dimensional structure of SPEs and the temporal evolution of these events in order to enhance SPE forecasts and warnings.<br><br>Presentations/Publications: First Year · Abstract and Poster “Preliminary MARIE Observations of Solar Particle Events” at the Solar, Heliospheric, and Interplanetary Environment Workshop, August, 2002, in Banff, Canada. · Abstract and talk “MARIE Observations of Solar Particle Events” at the 2003 Bioastronautics Investigators' Workshop, Galveston, Texas, January 2003 · Abstract and talk “MARIE Observations of Solar Particle Events” at the 14th Space Radiation Health Investigators’ Workshop, April, 2003.<br><br>Second Year · “Solar Proton Event Observations at Mars with MARIE,” B. Atwell, C. Zeitlin, F. Cucinotta, T. Cleghorn, P. Saganti, K. Lee, V. Andersen, L. Pinsky, R. Turner, 2003-01-2329, 33rd International Conference on Environmental Systems (ICES), July 2003. · Abstract and talk “MARIE Observations of Solar Particle Events” at the 15th Space Radiation Health Investigators’ Workshop, May, 2004.<br><br>Third Year · Invited talk, Chapman Conference on Solar Energetic Plasmas and Particles, “Space Weather Challenges Intrinsic to the Human Exploration of Space” 2-6 August 2004, Turku, Finland (paper submitted to AGU Monograph of proceedings) · Abstract and Poster, “MARIE Observations of Solar Particle Events” at the 2005 Bioastronautics Investigators' Workshop, Galveston, Texas, January 2005 · Abstract and talk “How Bad Can an SPE Be” at the 16th Space Radiation Health Investigators’ Workshop, May, 2005. |
| **Bibliography Type:** | Description: (Last Updated: 01/14/2008) |
| **Abstracts for Journals and Proceedings** | Turner R. "MARIE Observations of Solar Particle Events." 2005 Bioastronautics Investigators' Workshop, Galveston, Texas, January 2005.<br>2005 Bioastronautics Investigators' Workshop, 2005. , Jan-2005 |
| **Abstracts for Journals and Proceedings** | Turner R. "How Bad Can an SPE Be." 16th Space Radiation Health Investigators’ Workshop, May, 2005.<br>16th Space Radiation Health Investigators’ Workshop, 2005. , May-2005 |
| **Papers from Meeting Proceedings** | Turner R. "Space Weather Challenges Intrinsic to the Human Exploration of Space." Chapman Conference on Solar Energetic Plasmas and Particles, Turku, Finland, 2-6 August 2004.<br>Chapman Conference on Solar Energetic Plasmas and Particles, 2004. AGU monograph proceedings, in press. , Aug-2004 |